03.04: packages depens ✓

(torch 2.5.1 has built-in CUDNN 9.1.0)

test for ✓ | SVD, eigdecomp in numpy / scipy

contraction in opt_einsum

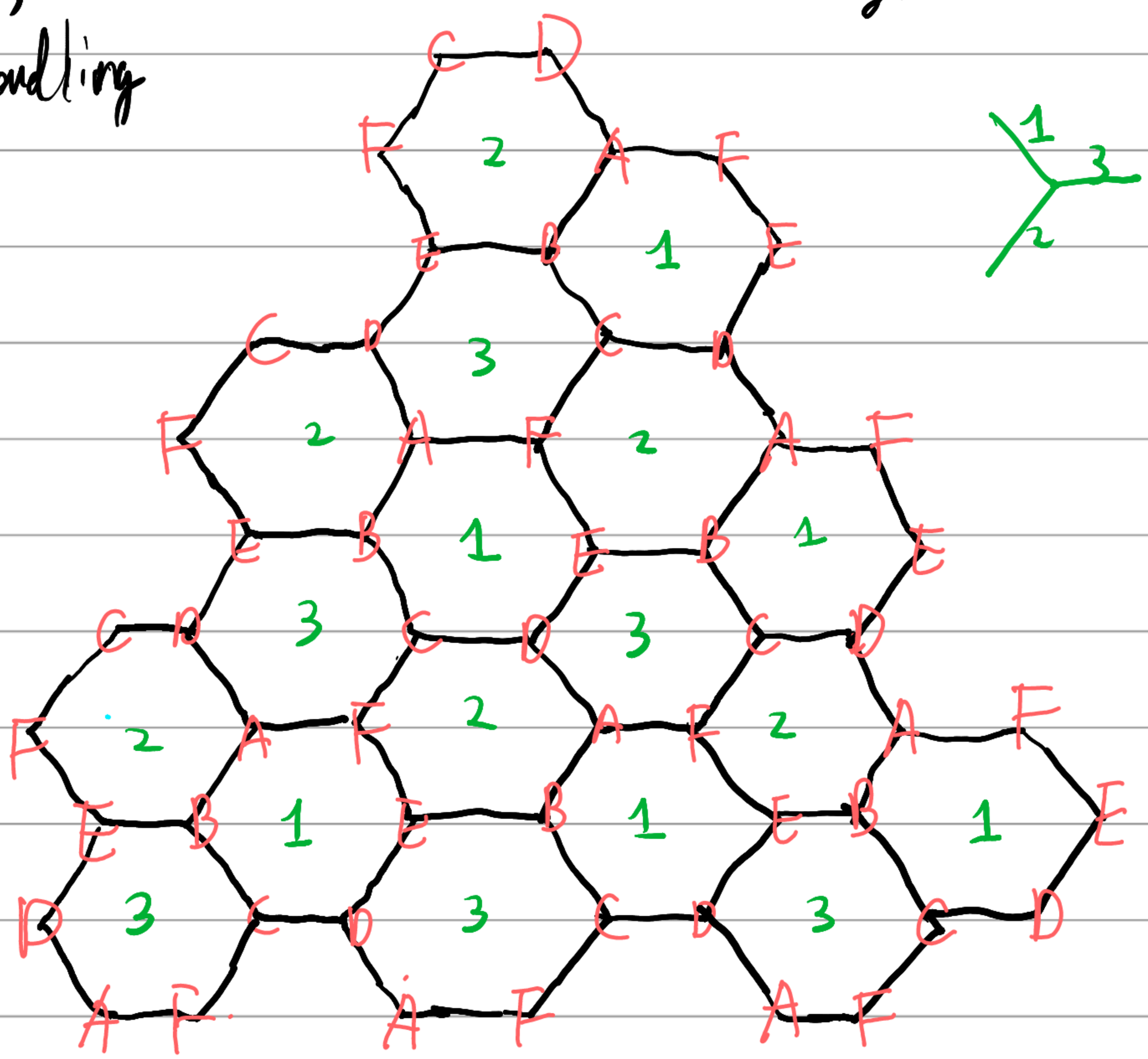
convection, GPU

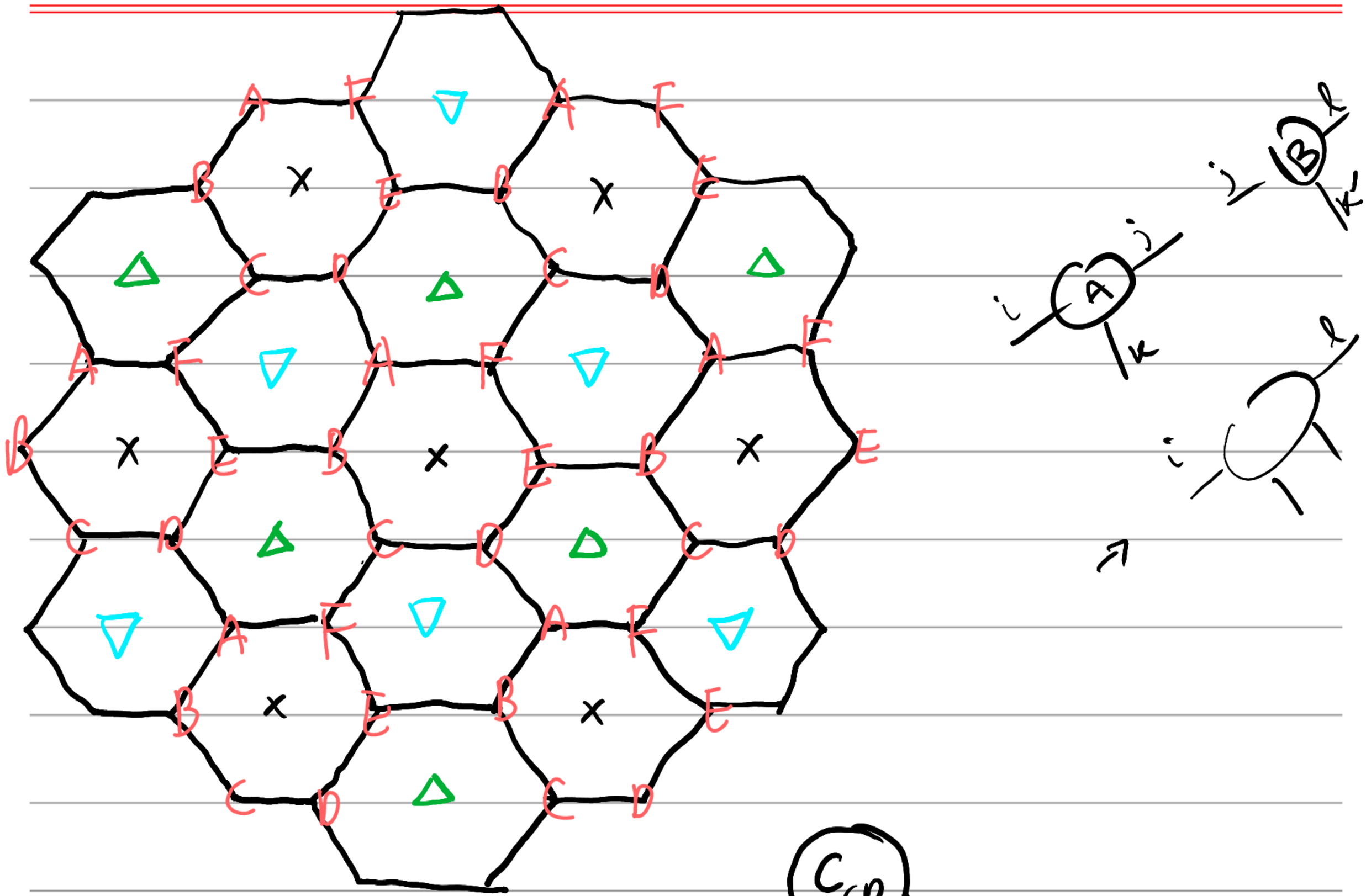
autograd { einsum, SVD, eigh } for torch

truncate SVD !!!

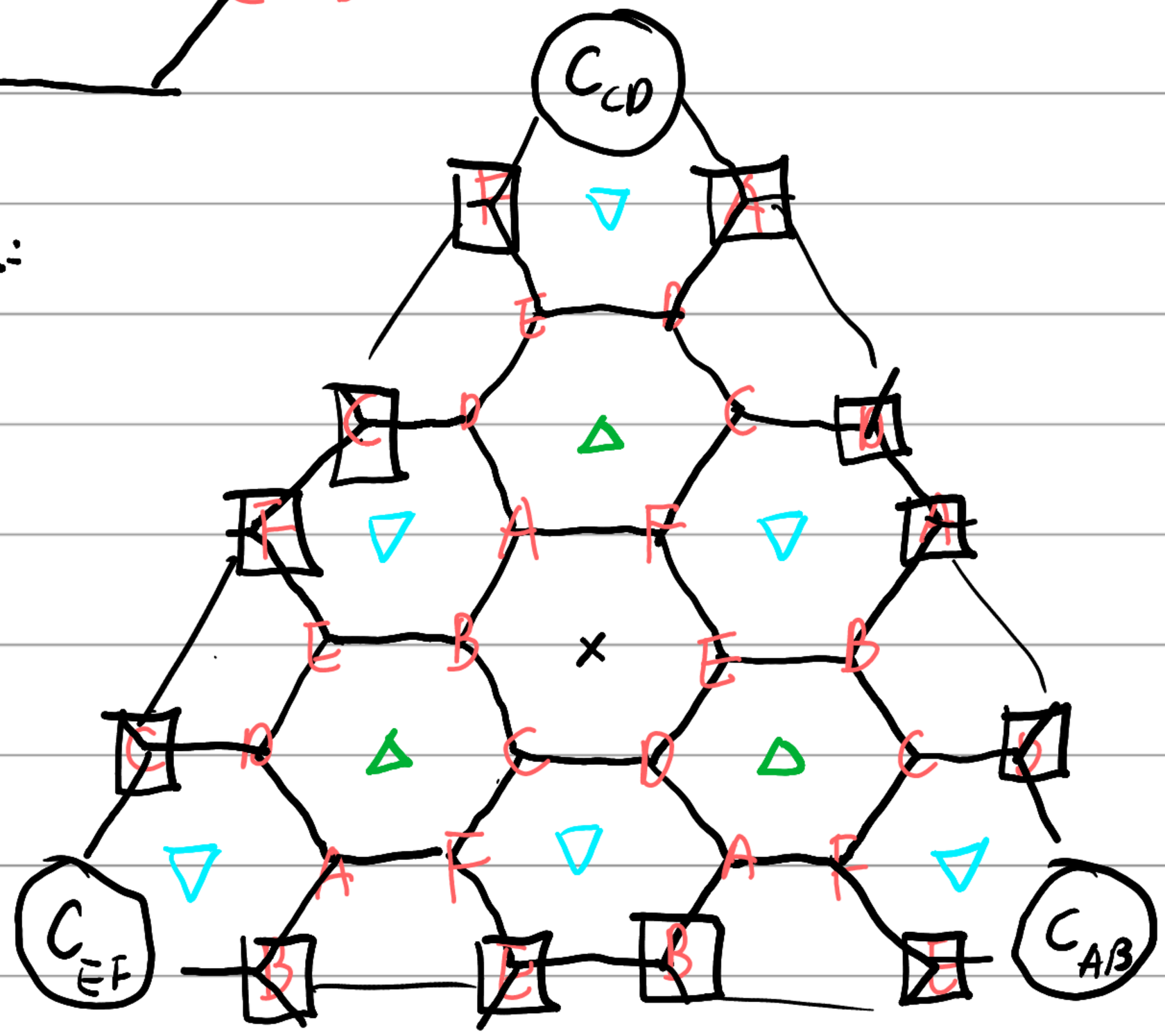
03.05

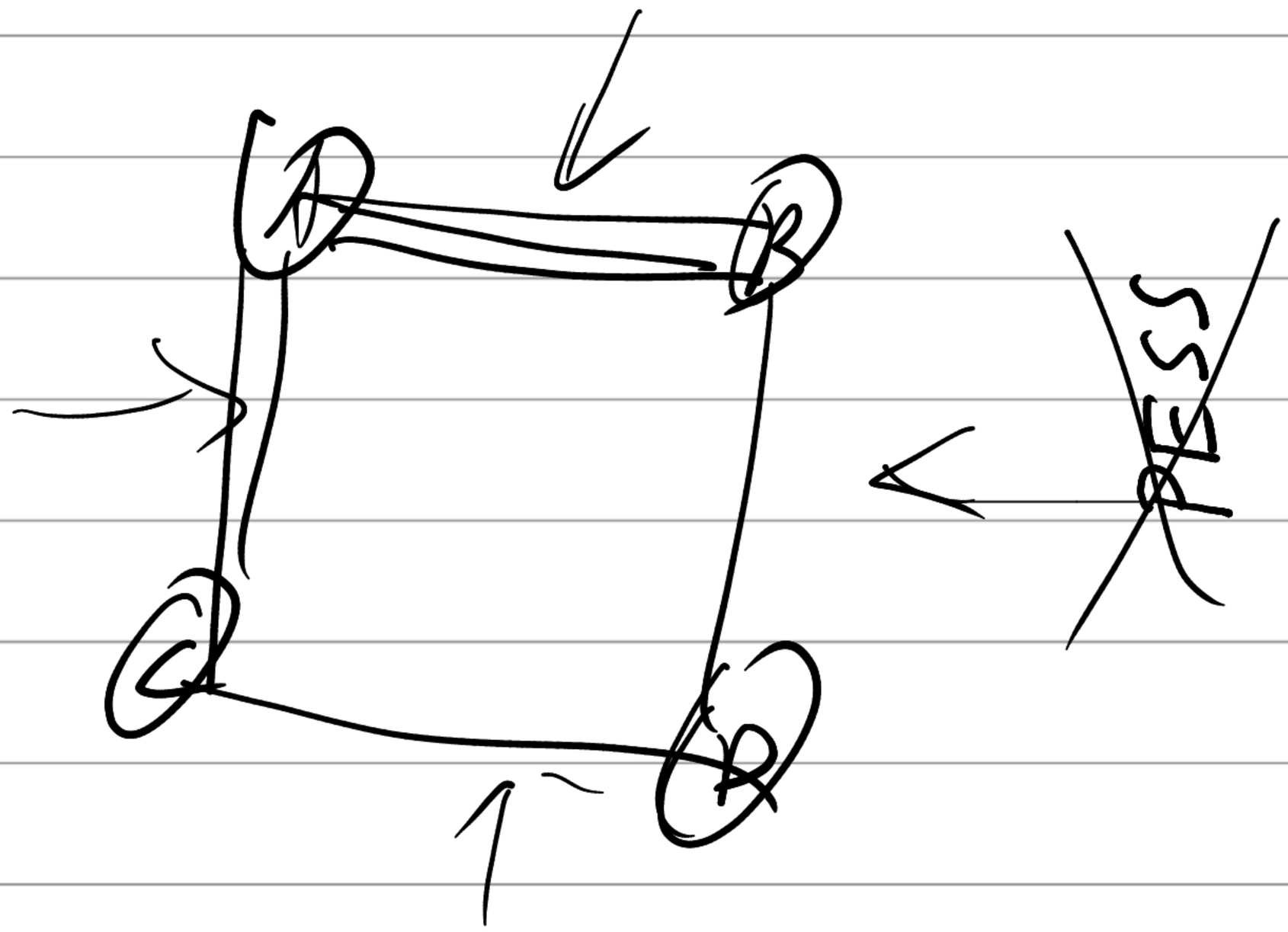
index handling





for center $\boxed{x}$:



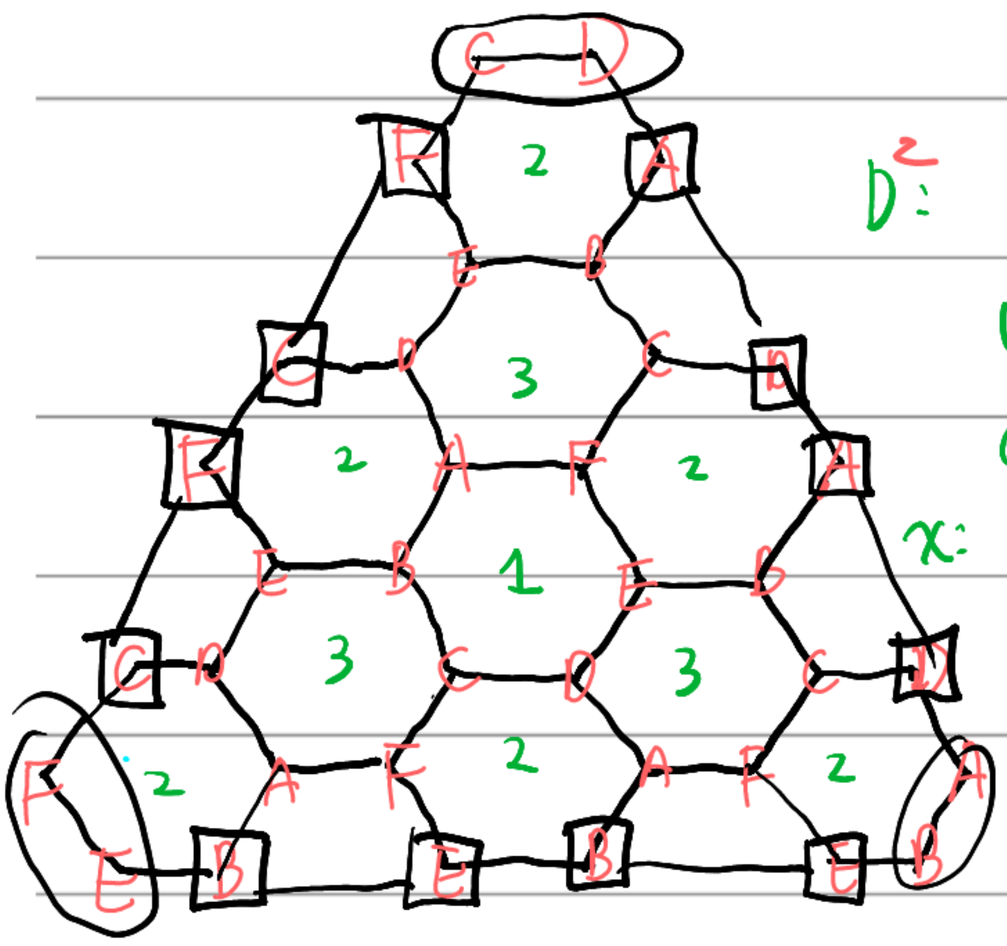


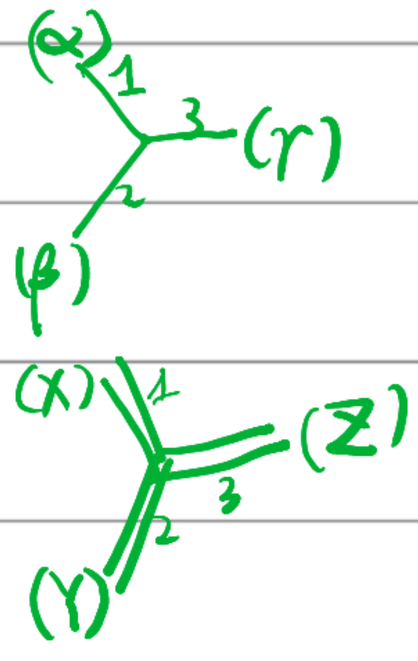
final Σ: $\sim 10^{-5 \sim -6}$

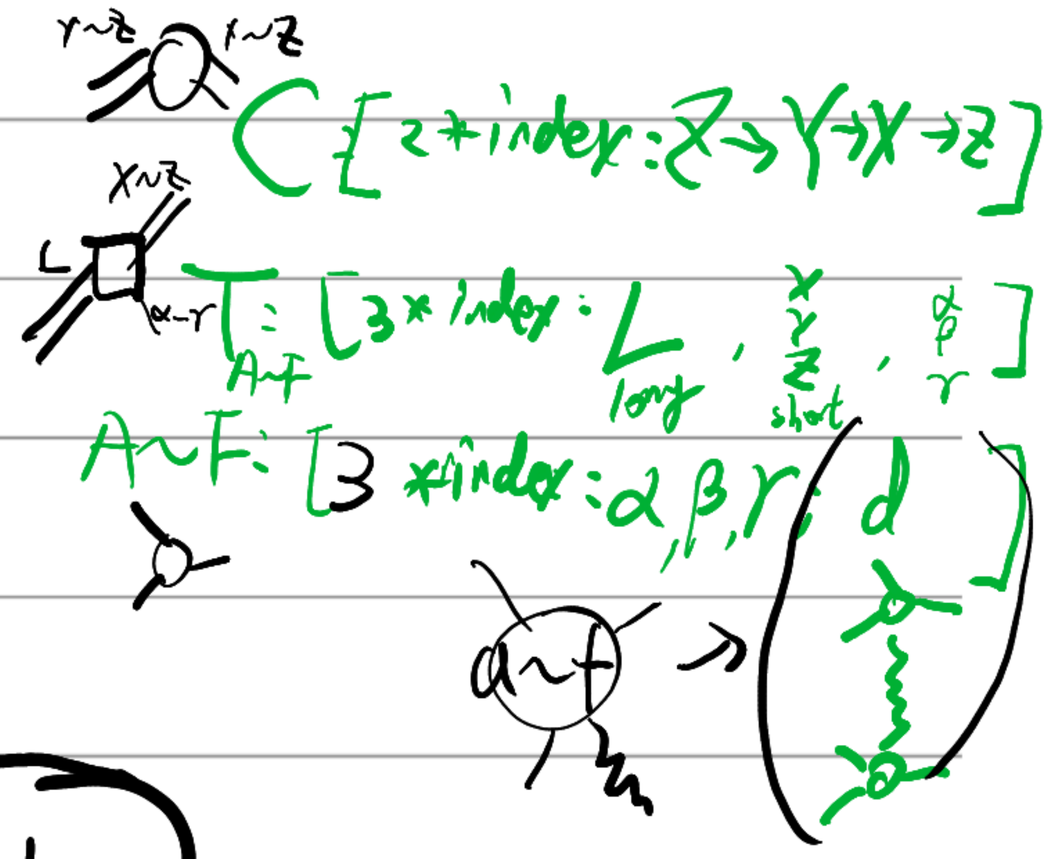
machine precision: float 32

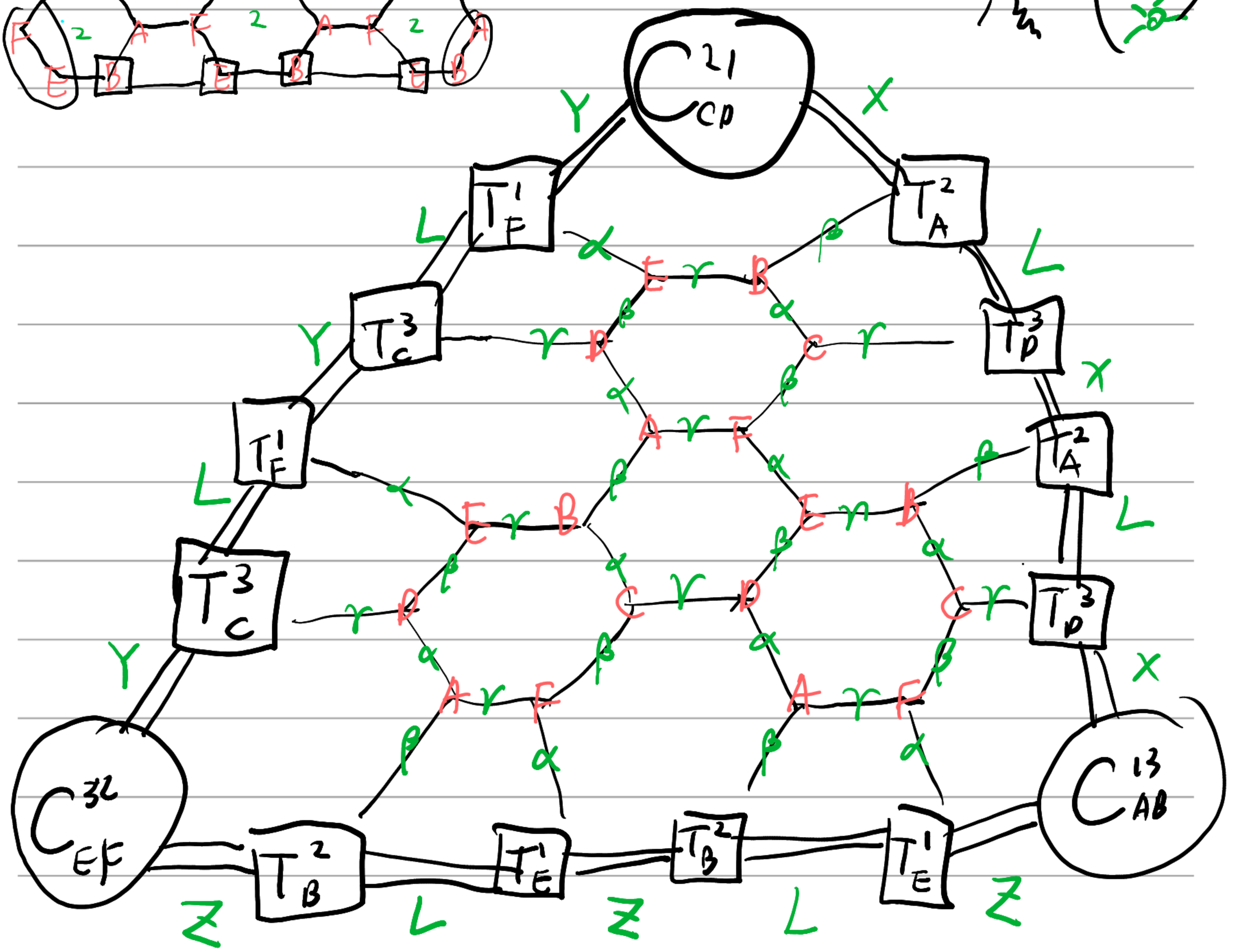
$\therefore$ random SVD 能达 $10^{-9} \sim 10^{-10}$

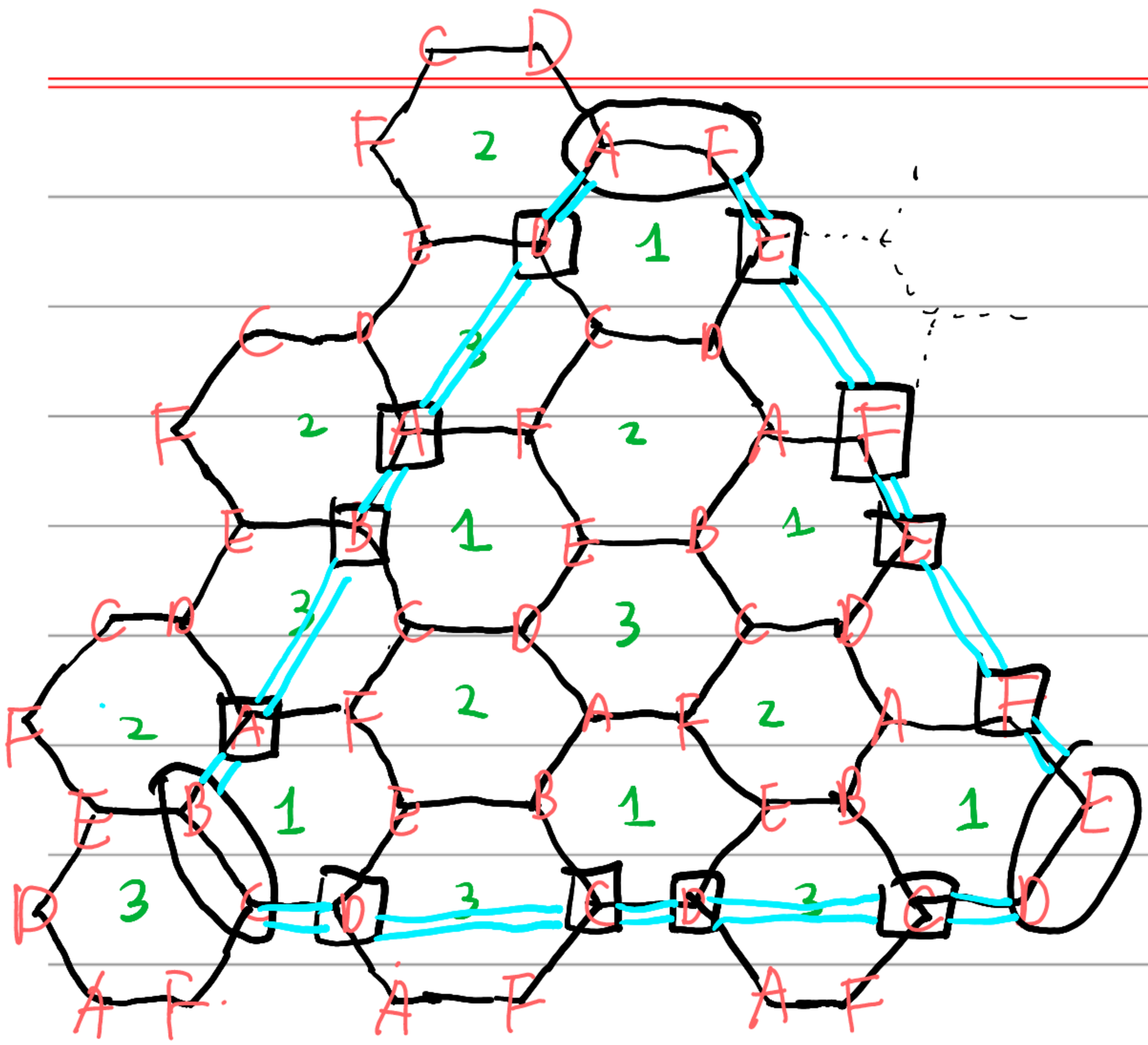
后面test, compared for 正常SVD
for CIMRG

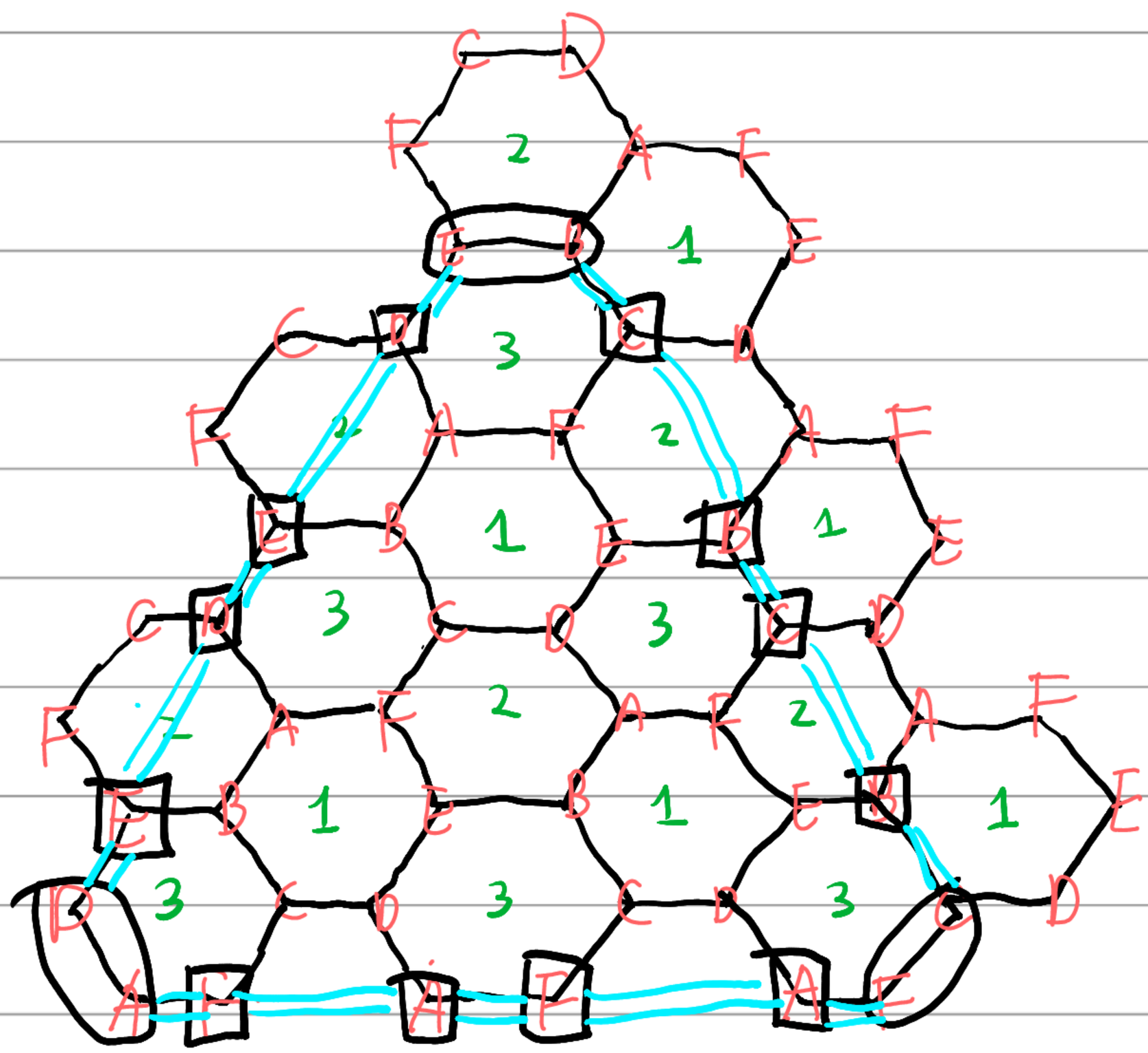



D:
x:


C [2*index: Z→Y→X→Z]
T: [3*index: L, X, α]
A~F: [3*index: α, β, γ]


C CD
C AB
C EF
T F
T A
T C
T D
T B
T E





C^{21}: CD → EB, CD → AF, EB → AF [YX] $\quad T^1 T^2$ [N, Z, αβ]

C^{32}: EF → AD, EF → CB, AD → CB [ZY] $\quad T^2 T^3$ [L, X, βγ]

C^{13}: AB → CF, AB → ED, CF → ED [XZ] $\quad T^3 T^1$ [M, Y, γα]

$C(X \times X)$ [Z→Y→X→Z *2]

$T(X \times X \times D^2)$ [Longer X, M Y, N Z, αβγ]

$A \sim F(D^2 \times D^2 \times D^2)$ [α, β, γ]

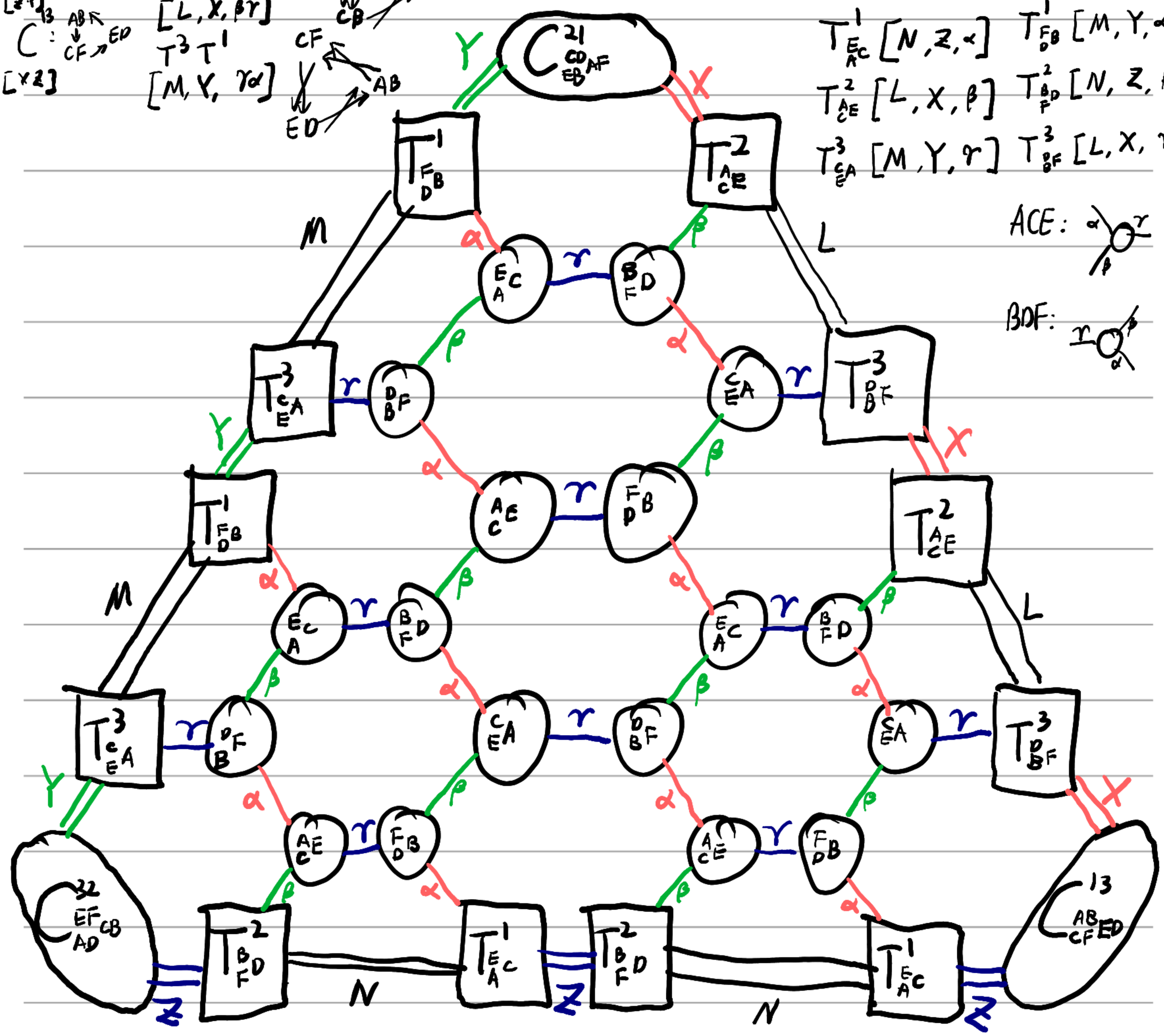
T^1_{EC} [N, Z, α] $\quad T^1_{FB}$ [M, Y, α]

T^2_{AE} [L, X, β] $\quad T^2_{BD}$ [N, Z, β]

T^3_{CA} [M, Y, γ] $\quad T^3_{BF}$ [L, X, γ]

ACE: α γ β

BDF: γ β α



C^{21}: CD → EB, CD → AF, EB → AF $\quad T^1_{EC} \to T^2_{BD} \to T^1_{EC} \to T^2_{BD} \to T^1_{EC} \to T^2_{BD} \to T^1_{EC}$

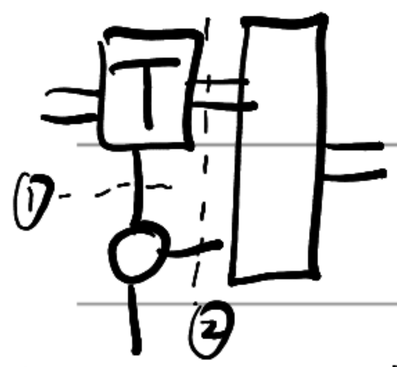
C^{32}: EF → AD, EF → CB, AD → CB $\quad T^2_{AE} \to T^3_{DF} \to T^2_{AE} \to T^3_{DF} \to T^2_{AE} \to T^3_{DF} \to T^2_{AE}$

C^{13}: AB → CF, AB → ED, CF → ED $\quad T^3_{CA} \to T^1_{FB} \to T^3_{CA} \to T^1_{FB} \to T^3_{CA} \to T^1_{FB} \to T^3_{CA}$

① center $B^{A\,F}_{C\,D\,E}$

max int. memory: $\chi^2(D^2)^2$

FLOPs: $\chi^3(D^2)^2 + \chi^2(D^2)^3$



independ:

max int. memory: $\chi^2(D^2)^2$

FLOPs: $\chi^2(D^2)^4 + \dots$

$[(0,2),(0,3),(1,2),(0,1)]$

reuse:

$\chi^3(D^2)^3 + \dots$

$[(3,4),(0,1)+3]$

$C^{21}_{CD},\ T^1_F,\ T^2_A,\ E,\ B$ $\xrightarrow{\text{MANUAL!!}}$ $C^{21\,(big)}_{EB}$

$YX,\ M\gamma\alpha,\ L\chi\beta,\ \alpha m\gamma,\ \ell\beta\gamma$

$M\,m\,L\,\ell \xrightarrow{\text{reshape}} (M*m,\ L*\ell)$

exact SVD

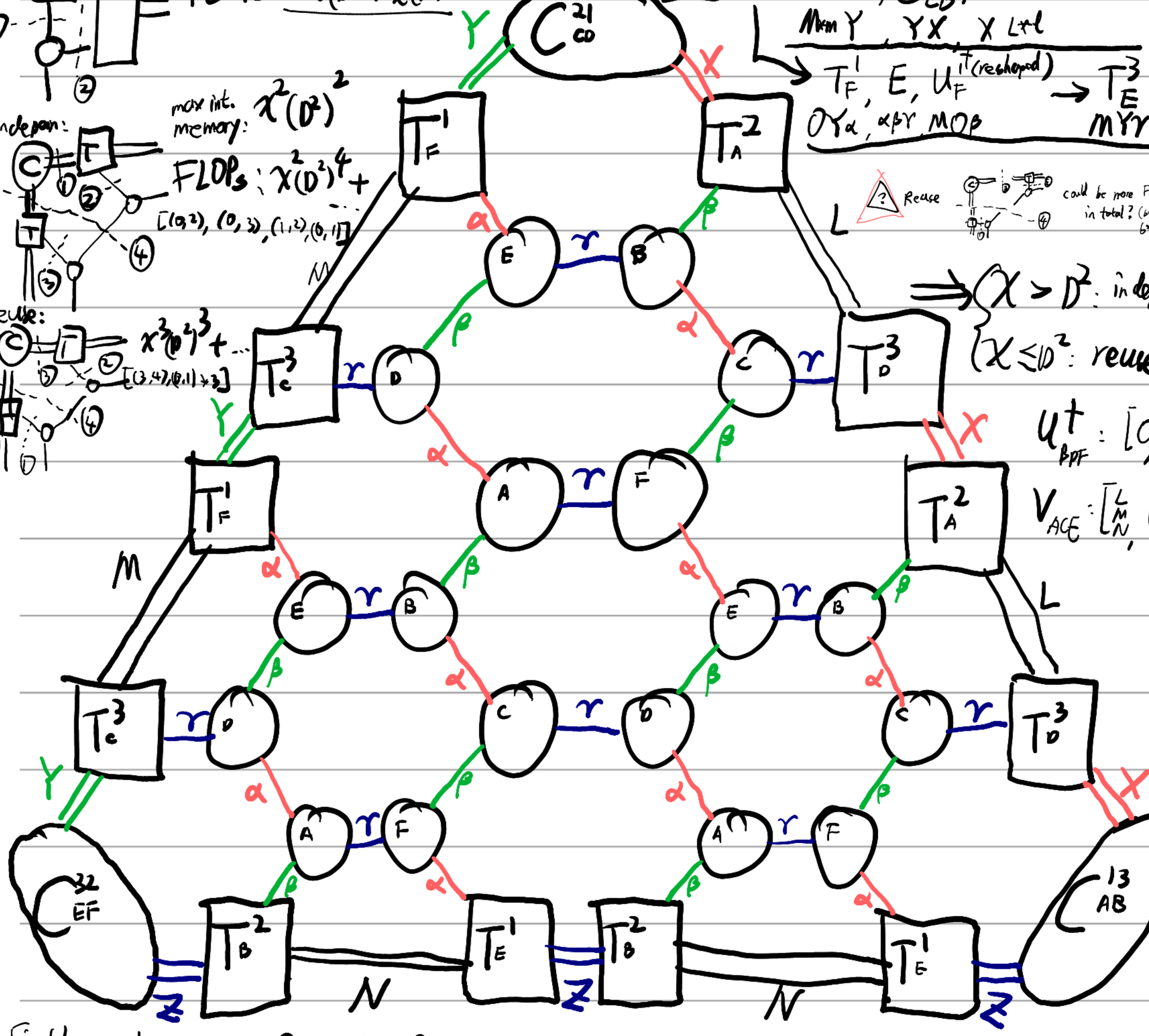
svd_lowrank

⚠ REUSE! U^{β}_C, C^{21}_{EB}, $U^{\beta\dagger}_B$

$M*m\,Y$, YX, $X\,L*\ell$

$T^1_F,\ E,\ U^{1\dagger(\text{reshaped})}_F \rightarrow T^3_E$

$O\gamma\alpha,\ \alpha\beta\gamma,\ MO\beta$ $\quad M\gamma\gamma$



⚠? Reuse

could be more FLOPs saving in total? (because of σ = T T T ?)

$\Rightarrow \begin{cases} \chi > D^2: \text{independent} \\ \chi \le D^2: \text{reuse} \end{cases}$

$U^{\dagger}_{BPF}: [O, \alpha\beta\gamma, MN]$

$V_{ACE}: [LMN, O, \alpha\beta\gamma]$

Firstly, all "independ" & exact SVD.

$C^{21}_{CD},\ T^1_F,\ T^2_A,\ E,B$ $\xrightarrow{[(0,2),(0,3),(1,2),(0,1)]}$ $C^{21\,(big)}_{EB}$ $\xrightarrow{\text{reshape}}$ $C^{21\,(mat)}_{EB}$ $\xrightarrow[[:\chi]]{\text{SVD}}$ $U^3_C,\ C^{21}_{EB},\ U^{3\dagger}_D$ $\quad T^1_F,\ E,\ U^{1\dagger(\text{reshape})}_F \rightarrow T^3_E$ $[(0,1),(0,1)]$

$YX,\ M\gamma\alpha,\ L\chi\beta,\ \alpha m\gamma,\ \ell\beta\gamma$ $\quad M\,m\,L\,\ell$ $\quad M*m,\ L*\ell$ $\quad M*m,\ Y$ $\quad YX$ $\quad X,\ L*\ell$ $\quad O\gamma\alpha,\ \alpha\beta\gamma,\ O\beta M \rightarrow M\gamma\gamma$

$C^{32}_{EF}\ T^2_B\ T^3_C\ A\ D$ $\rightarrow$ $C^{32\,(big)\rightarrow(mat)}_{AD}$ $\xrightarrow{\text{SVD}}$ $U^1_E,\ C^{32}_{AD},\ U^{1\dagger}_F$ $\quad T^2_A,\ B,\ V^2_A \rightarrow T^3_B$ $\quad T^2_B\ A\ U^{2\dagger}_B \rightarrow T^1_A$

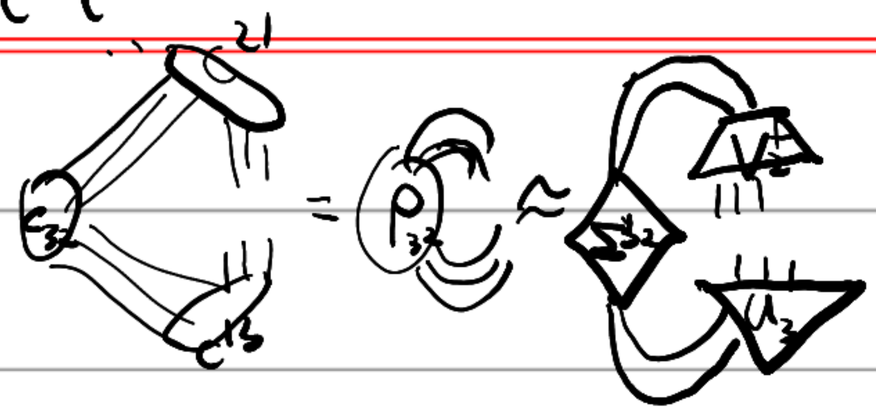
$Z\chi,\ NZ\beta,\ M\gamma\gamma,\ \alpha\beta n\ \alpha m\gamma$ $\quad N*n,\ M*m$ $\quad Nn,\ Z$ $\quad ZY$ $\quad Y,\ Mm$ $\quad O\chi\gamma\ \alpha\beta\gamma$ $\quad MO\beta$ $\quad O\chi\beta,\ \alpha\beta\gamma,\ LO\alpha$ $\quad L\chi\gamma$ $\quad O Z\beta\ \alpha\beta\gamma\ O\gamma N\ NZ\alpha$

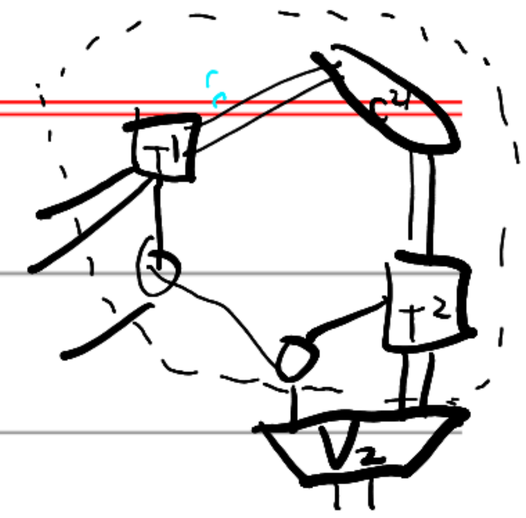
$T^3_C\ D\ V^3_C \rightarrow T^1_D$ $\quad M\gamma\alpha$

$C^{13}_{AB}\ T^3_D\ T^1_E\ C\ F$ $\rightarrow$ $C^{13\,(big)\rightarrow(mat)}_{CF}$ $\xrightarrow{\text{SVD}}$ $U^2_A,\ C^{13}_{CF},\ U^{2\dagger}_B$ $\quad T^3_D\ C\ U^{3\dagger}_D \rightarrow T^2_C$ $\quad T^1_E\ F\ V^1_E \rightarrow T^2_F$

XZ $\quad L\chi\gamma\ NZ\alpha\ \ell\beta\gamma\ \alpha\beta n$ $\quad L\ell,\ Nn$ $\quad L\ell,\ X$ $\quad XZ$ $\quad Z,\ Nn$ $\quad O\chi\gamma$ $\quad \alpha\beta\gamma$ $\quad O\alpha L$ $\quad L\chi\beta$ $\quad O Z\alpha\ \alpha\beta\gamma\ N O\gamma\ NZ\beta$

$C^{13}C^{32}C^{21}$





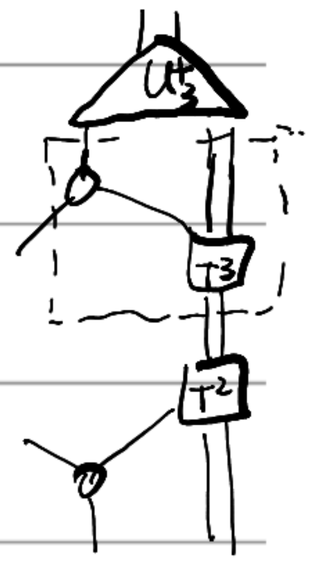
② center $\begin{matrix} & C & D \\ F & & A \\ & E & B \end{matrix}$

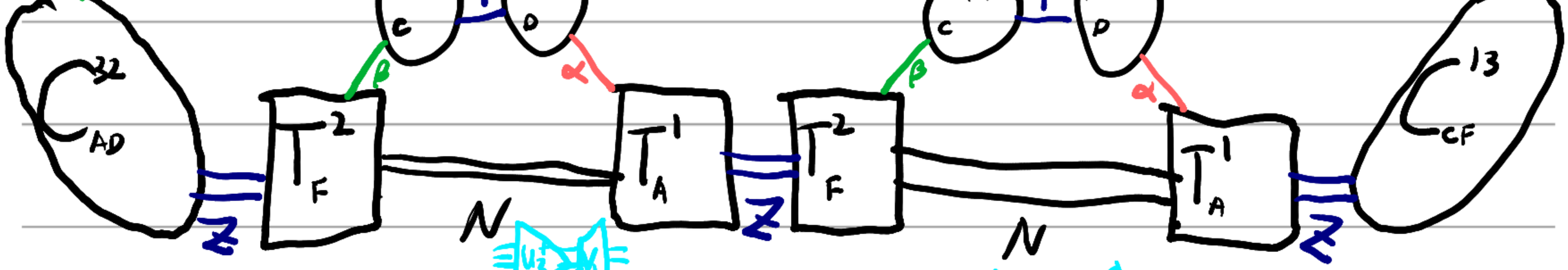
$\rho_{32} = \begin{matrix} C_{13}C_{32}C_{21} \\ C_{21}C_{13}C_{32} \\ C_{32}C_{21}C_{13} \end{matrix} \rightsquigarrow \begin{matrix} U_3 & \Sigma_{32} & V_2^\dagger \\ 1 & 13 & 3 \\ 2 & 21 & 1 \end{matrix}$

if ρ_s herm

$U_3 = V_2$

$\begin{matrix} 1 & 3 \\ 2 & 1 \end{matrix}$



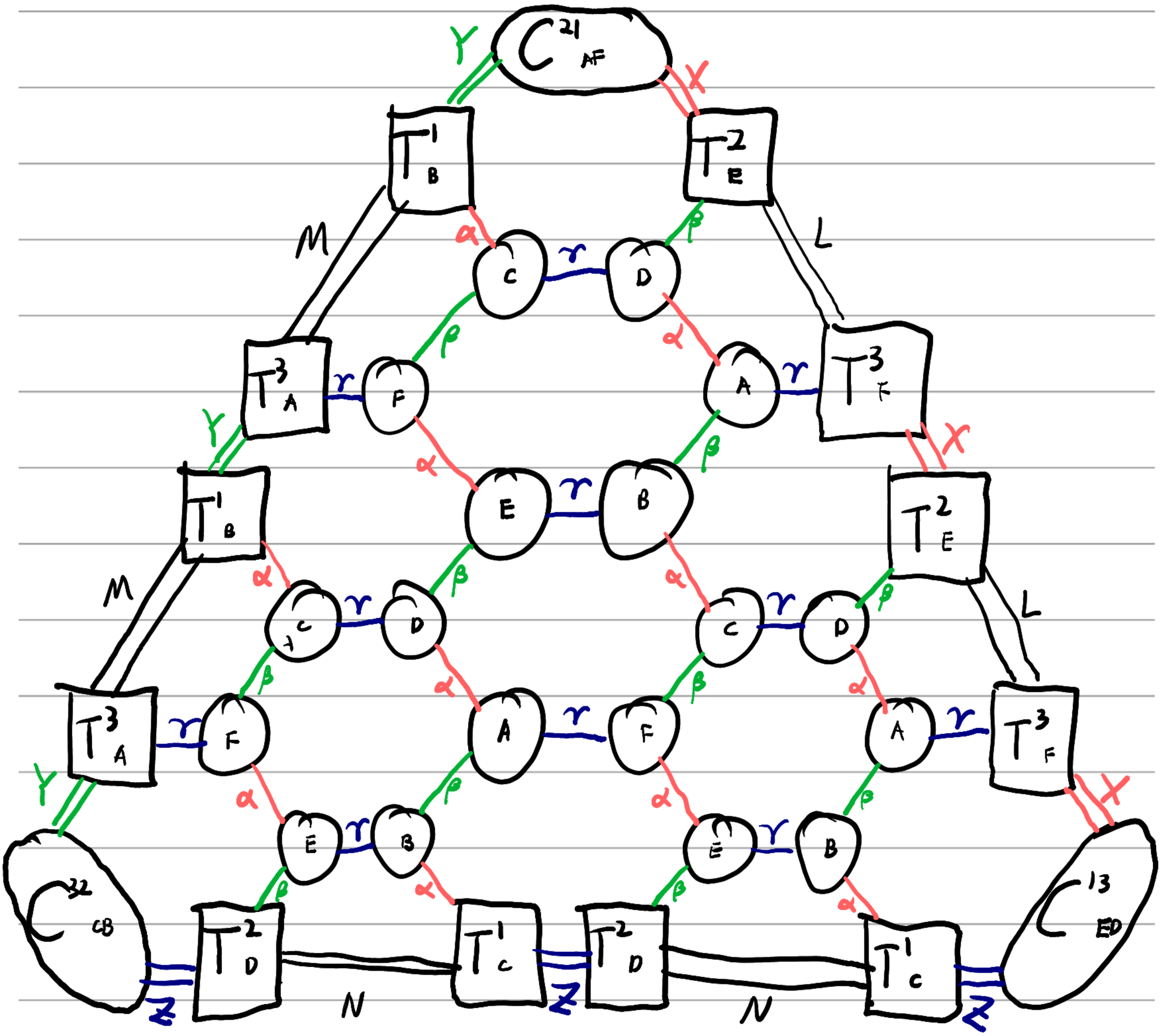


C^{21}: CD, EB, AF — $T^1 T^2$ — $[N, Z, \alpha\beta]$

C^{32}: EF, AD, CB — $T^2 T^3$ — $[L, X, \beta\gamma]$

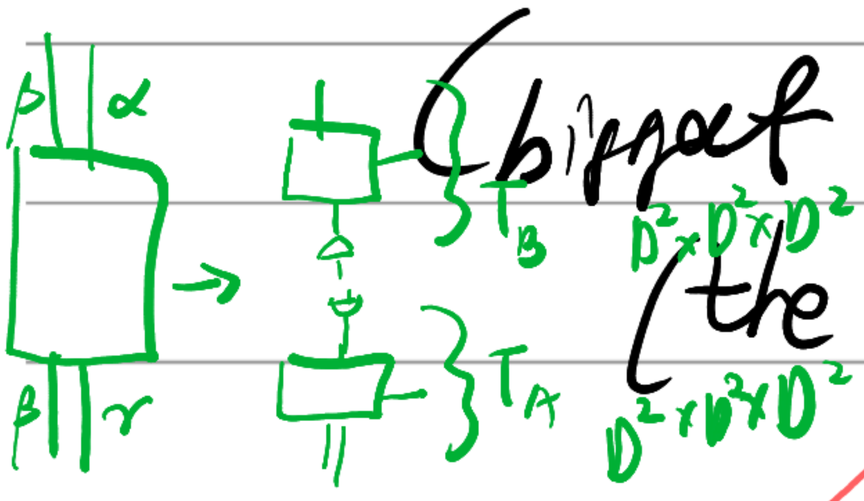
C^{13}: AB, CF, ED — $T^3 T^1$ — $[M, Y, \gamma\alpha]$

3 center $\begin{matrix} & E & B \\ D & & & C \\ & A & F \end{matrix}$



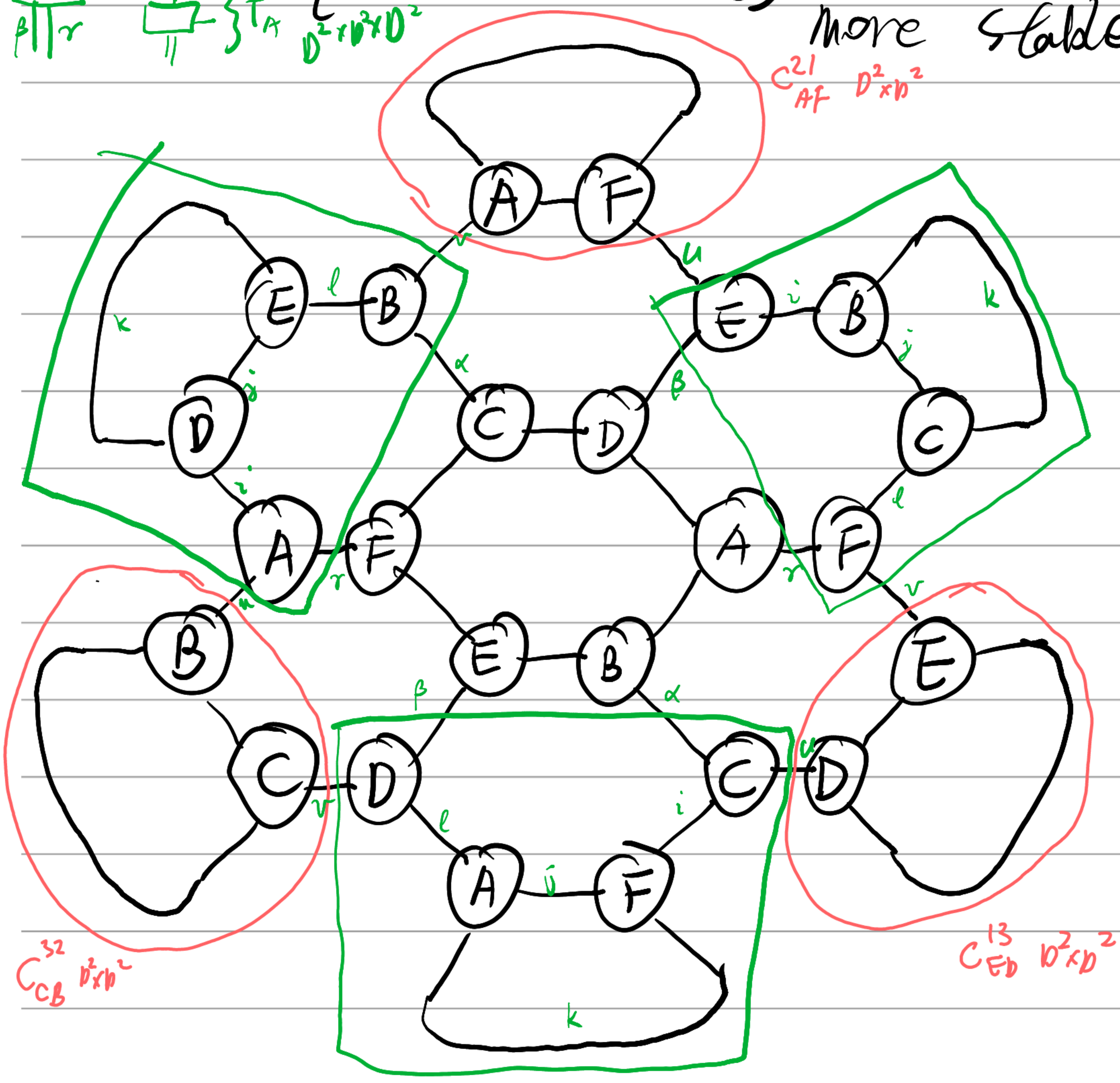
track → discarded mesh

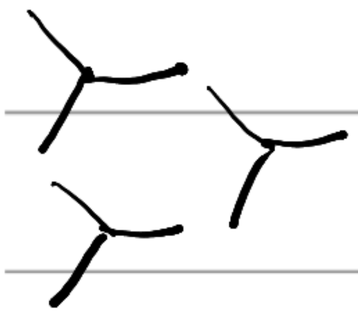
good guess makes beginning SVD more stable!



(biggest truncation)

(the ε related to x!)



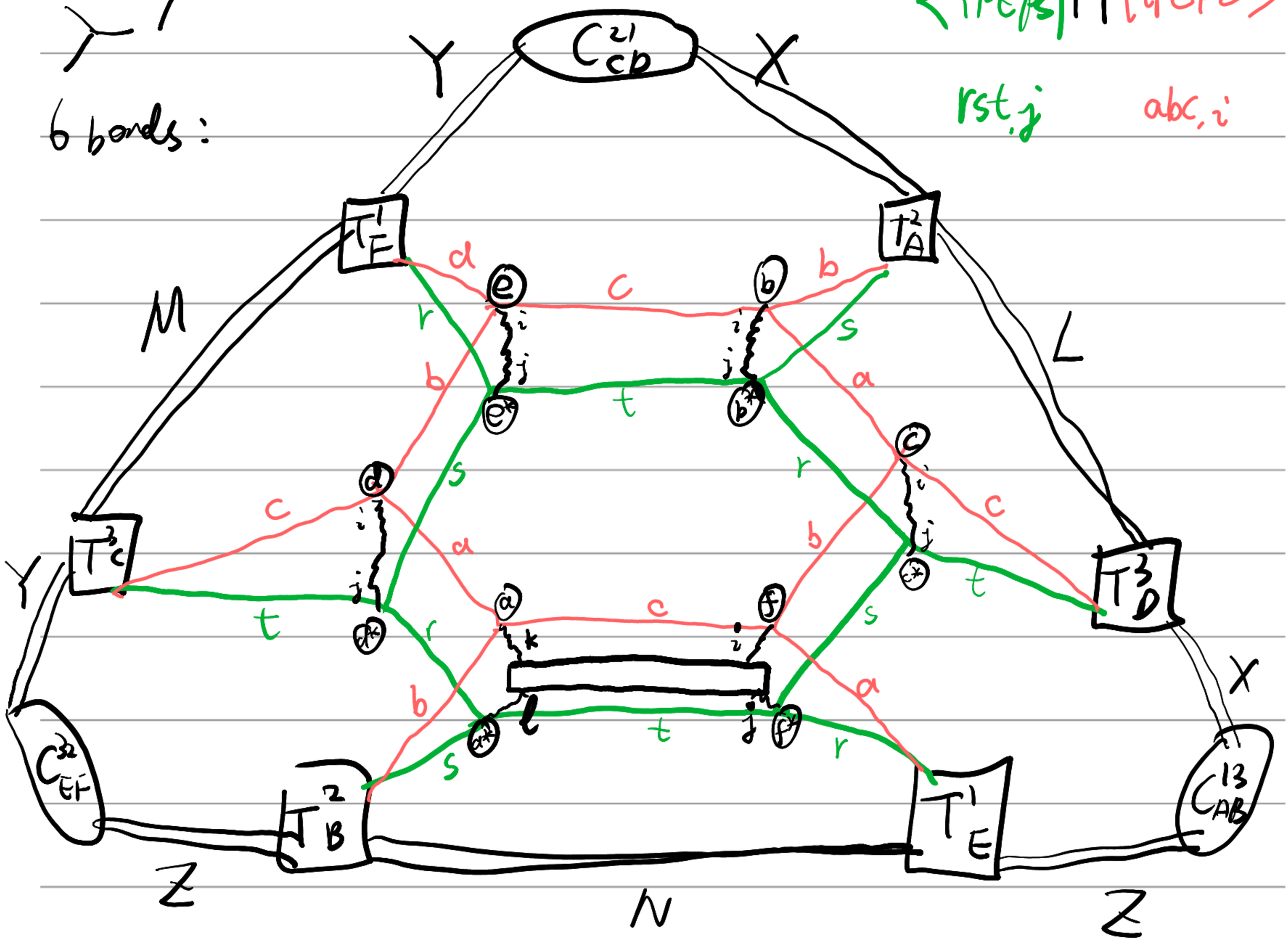


= 6+3 bonds

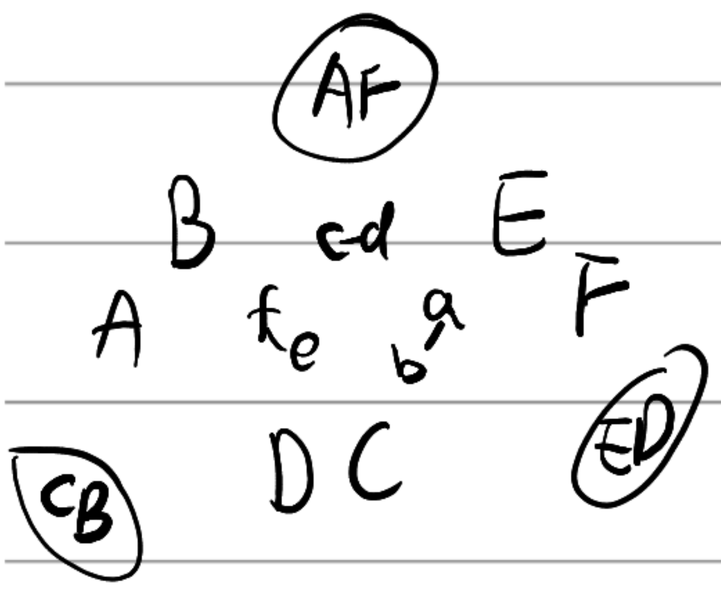
$\langle iPEPS | H | iPEPS \rangle$

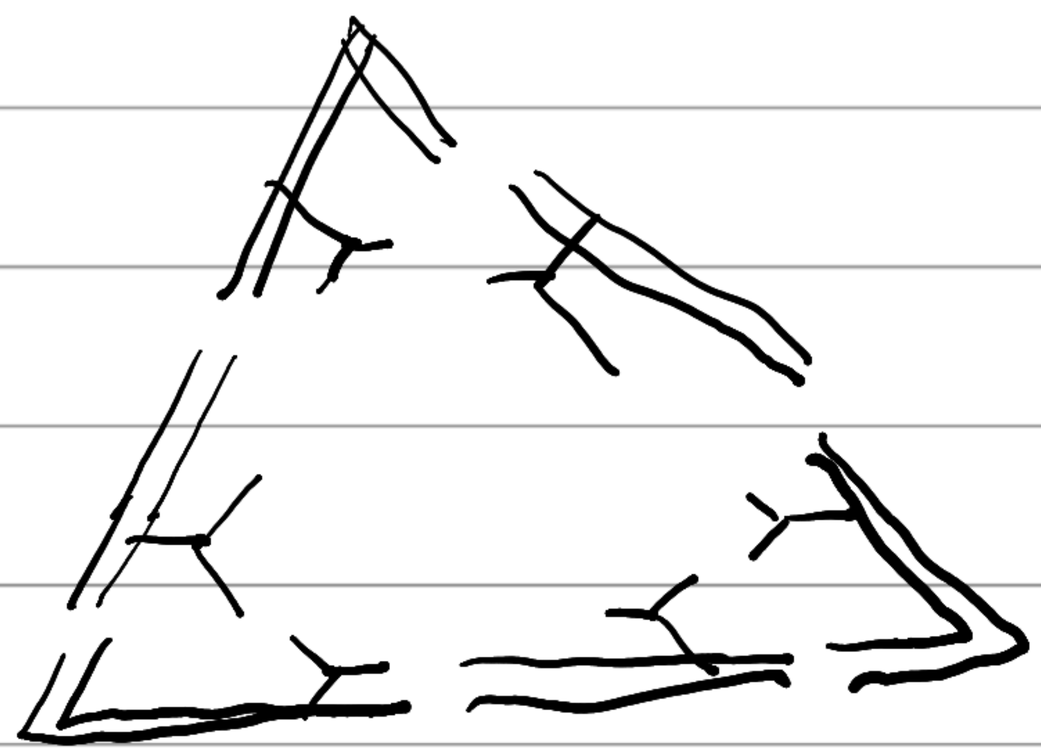
rst, j abc, i

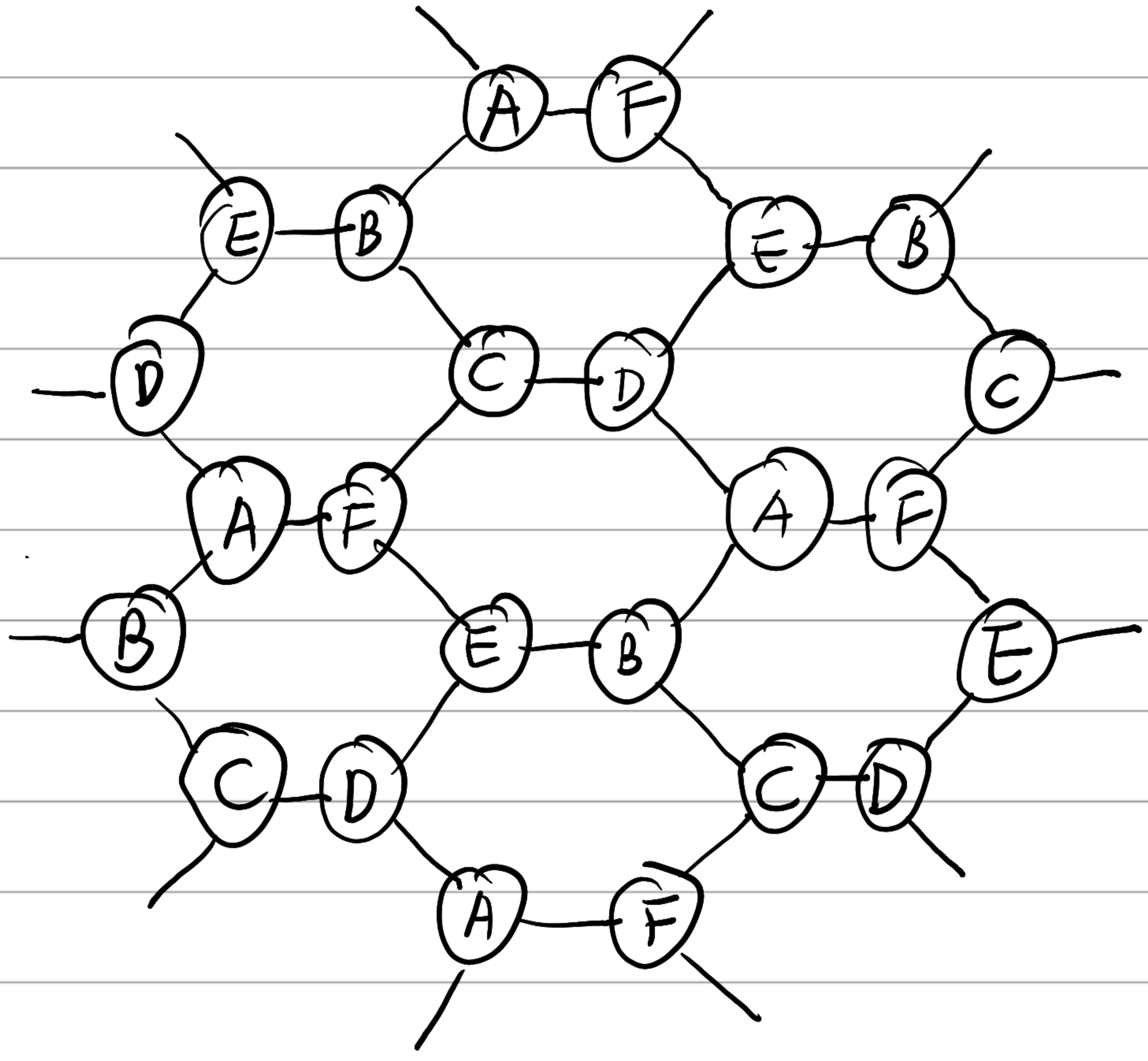
6 bonds:



other 3 bonds:






A
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